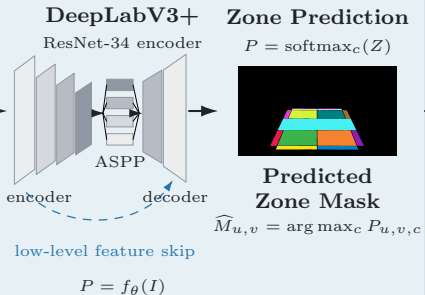


Input Frame

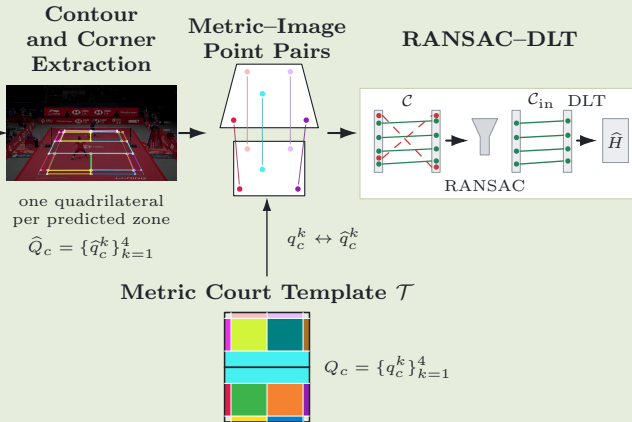


$$I \in \mathbb{R}^{H \times W \times 3}$$

Semantic Segmentation



Zone-Based Geometric Registration



Template Projection



Registered Court

projected zones and court lines

$$\hat{x} \sim \hat{H}X$$